■ Higher Flame Field Guide

Metabolic Flexibility, Mitochondrial Respiration Voltage & Fuel Switching

Metabolic health is the capacity to seamlessly transition between burning glucose and fatty acids. High mitochondrial membrane potential (ΔΨm) sustains cellular ATP generation.

Core Biophysics Principles

1. Mitochondrial Membrane Potential (ΔΨm)
The inner mitochondrial membrane maintains a voltage gradient (-140mV to -180mV) that drives the electron transport chain.

2. Substrate Switching Dynamics
Healthy metabolism easily toggles between carbohydrate oxidation post-meal and beta-oxidation during fasting or exertion.

3. AMPK & sirtuin Pathway Co-Activation
Energy sensors (AMPK, SIRT1, SIRT3) orchestrate mitochondrial biogenesis and GLUT4 glucose transporter translocation.

Key Protocol Elements & Interventions

• Mitochondrial Co-Factors (CoQ10 Ubiquinol, Alpha-Lipoic Acid, PQQ)

• AMPK Activators (Berberine HCl, Gynostemma Pentaphyllum, D-Chiro-Inositol)

• NAD+ Precursors (Nicotinamide Riboside / NMN Support)

• Cold Exposure & Non-Shivering Thermogenesis Protocols

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